Angular Momentum and Rolling

Building Models to Describe our World

Rotational Energy, Momentum and RWS

Outline

Angular and Rolling Motion

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Rotational Energy, Momentum and RWS

Quick review: Rotational dynamics

- Torque on a body **about an axis**:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

- Rotational equivalent of Newton's Second Law:

$$\sum \vec{\tau}^{ext} = I\vec{\alpha}$$

- Moment of inertia of an object **about an axis**:

$$I = \sum m_i r_i^2 = \int_0^M r^2 dm$$

- We describe the motion of a rotating object using kinematic quantities (position, velocity, acceleration):

$$\omega = \frac{d\theta}{dt} \qquad \alpha = \frac{d\omega}{dt}$$

- In order to describe about which axis these refer to, we use the right hand rule for axial vectors to define directions for corresponding vectors:

$$\vec{\omega}, \vec{\alpha}$$

Why introduce rotational stuff?

- In the last few chapters, we have expanded our ability to build models from describing particles to “solid objects”.
- Newton’s Second Law only describes a point particle:

$$\sum \vec{F} = m\vec{a}$$

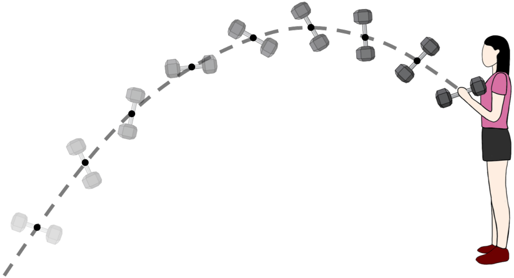
- By considering solid objects as being made of many point particles, we can now describe the motion of on an object as a whole:
  - How its centre of mass moves:

$$\sum \vec{F}^{ext} = M\vec{a}_{CM}$$

  - How the object rotates about the centre of mass:

$$\sum \vec{\tau}^{ext} = I\vec{\alpha}$$

Motion, in general



The motion of the dumbbell appears complicated, but is just a combination of motion of the CM and rotation about the CM.

Think of the motion of the dumbbell as the combination of:

1. Motion of the CM according to external forces (gravity $\rightarrow$ parabolic motion).
2. Uniform circular motion about the CM (no external torque).

Energy of an object that can rotate

- It takes energy to accelerate the centre of mass. We call this "translational kinetic energy".

$$K_{trans} = \frac{1}{2} M v_{CM}^2$$

- It takes energy to make the object rotate. We call this "**rotational kinetic energy**".

$$K_{rot} = \frac{1}{2} I \omega^2$$

- If an object's CM is moving, and the object is rotating, its total kinetic energy is the sum of these 2 (and **we take the axis of rotation through the CM**):

$$\boxed{K_{tot} = \frac{1}{2} M v_{CM}^2 + \frac{1}{2} I \omega^2}$$

Work-Energy theorem

- It takes work to give the object translational kinetic energy:

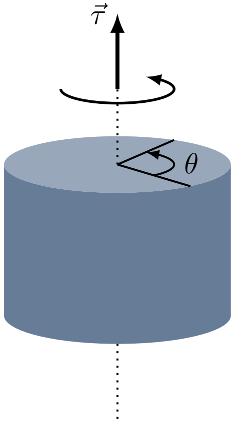
$$W^{net} = \int \vec{F}^{net} \cdot d\vec{l} = \Delta K_{trans}$$

The net force is exerted over a certain distance/displacement.

- It takes work to give the object rotational kinetic energy:

$$W^{net} = \int \vec{\tau}^{net} \cdot d\vec{\theta} = \Delta K_{rot}$$

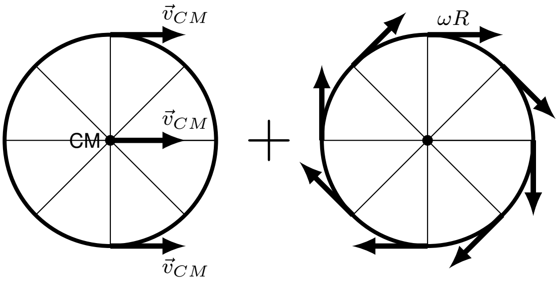
The net torque is exerted over a certain angular displacement.



Work done by torque over an angular displacement.

Rolling motion

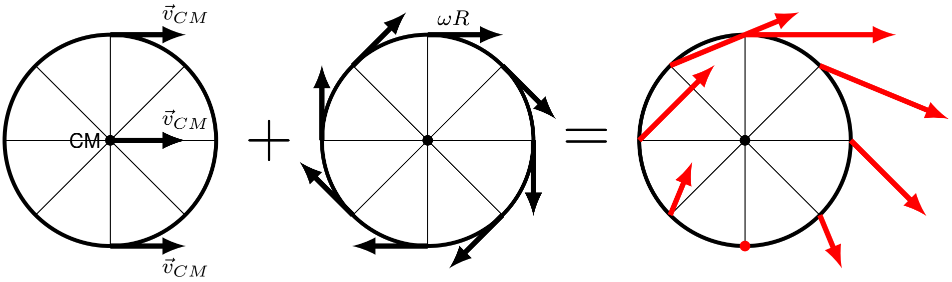
- A specific combination of translational and rotational motion.



Rolling motion.

Rolling motion

- A specific combination of translational and rotational motion.
- The velocity of any point with respect to the ground is the sum of:
  - The velocity of the CM.
  - The velocity of the point relative to the CM (with speed given by $v = \omega R$).

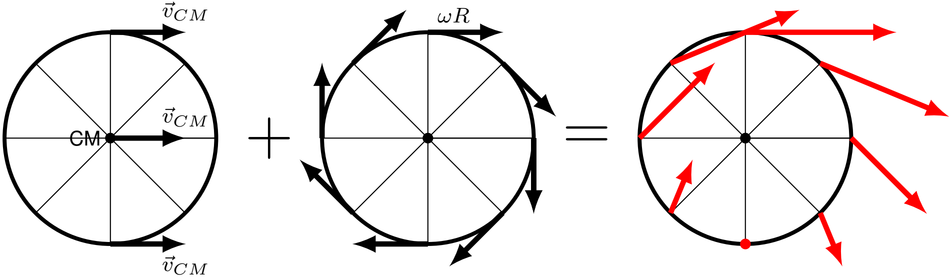


Rolling motion. In rolling without slipping, the point of contact must be at rest relative to the ground.

Rolling motion

- A specific combination of translational and rotational motion.
- The velocity of any point with respect to the ground is the sum of:
  - The velocity of the CM.
  - The velocity of the point relative to the CM (with speed given by $v = \omega R$).
- Can have rolling with or without slipping. If **without** slipping:

$$\boxed{v_{CM} = \omega R}$$



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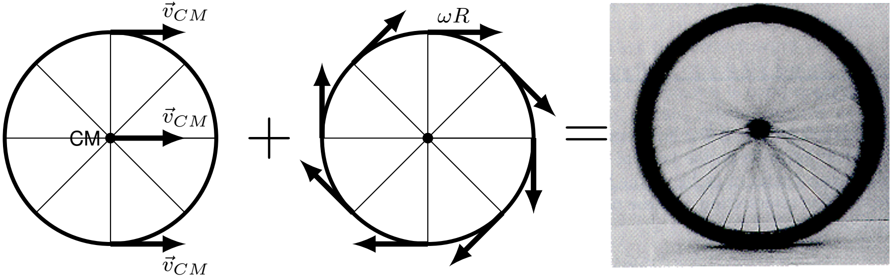

Rolling motion. In rolling without slipping, the point of contact must be at rest relative to the ground.

Rolling Without Slipping

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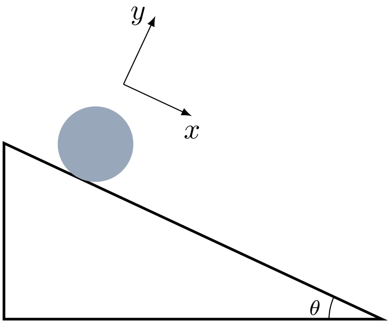
Modeling rolling without slipping

Group question

1. Draw the forces on the cylinder.
2. Write the equations of motion:
    - Sum of the forces in the x direction:
    - Sum of the forces in the y direction:
    - Sum of the torques about the CM:



A cylinder rolling down an incline.

Modeling rolling without slipping

1. Draw the forces on the cylinder.
2. Write the equations of motion:
   - Sum of the forces in the x direction:

$$\sum F_x = Mg\sin\theta - f_s = Ma_{CM}$$

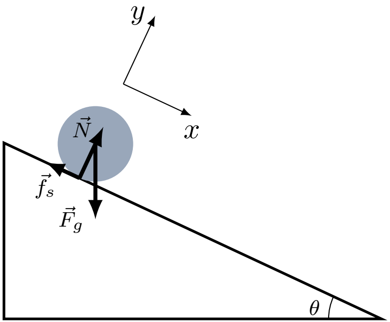
   - Sum of the forces in the y direction:

$$\sum F_y = N - Mg\cos\theta = 0$$

   - Sum of the torques about the CM:

$$\sum \tau_z = -f_s R = \frac{1}{2}MR^2\alpha$$

3. Can now solve for a_{CM}, f_s, N...



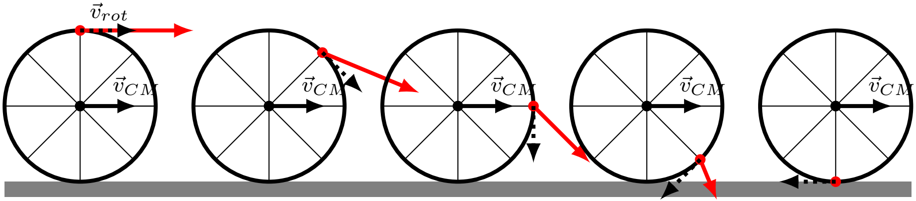
A cylinder rolling down an incline. If the incline is too steep, the force of static friction cannot be large enough to prevent slipping.

For rolling without slipping:

$$\boxed{a_{CM} = \alpha R}$$

The instantaneous axis of rotation

- In rolling **without** slipping, one can choose the instantaneous axis of rotation (through the point of contact with the ground) instead of the CM to build a model.
- About the instantaneous axis (fixed to the ground, so inertial frame of reference), the **motion is rotation only**, not the sum of rotation and translation
- The angular velocity is the same about the CM or about the instantaneous axis!



In rolling without slipping any point on the wheel can be modeled as rotating about the point of contact with the ground.

The instantaneous axis of rotation

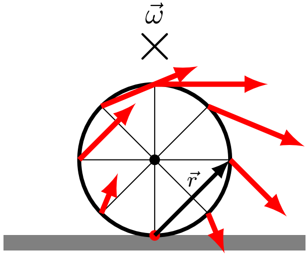
- About the instantaneous axis of rotation, each point is at a different distance from the axis.
- The velocity vector for any point is perpendicular to $\vec{r}$ and to $\vec{\omega}$. It is given by:

$$\vec{v} = \vec{\omega} \times \vec{r}$$

- For example, the speed of the CM is given by:

$$v_{CM} = ||\vec{\omega} \times \vec{r}_{CM}|| = \omega R$$

Which is the speed of the CM relative to the ground. **The angular velocity must be the same about the CM and about the instantaneous axis**.



About the instantaneous axis of rotation, all points appear to rotate around the axis with the same angular velocity.

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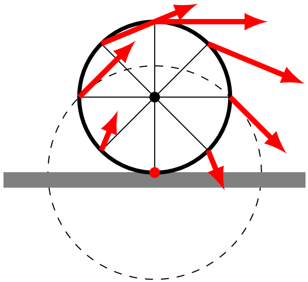


About the instantaneous axis of rotation, all points appear to rotate around the axis with the same angular velocity.

Modeling rolling without slipping about the instantaneous axis

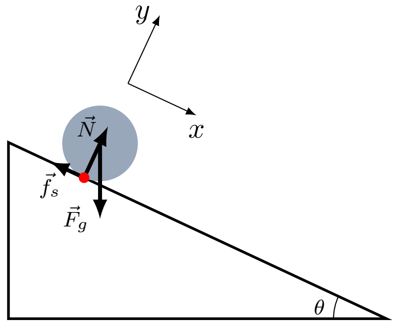
1. Draw the forces on the cylinder.
2. Write the equations of motion:
   - Sum of the forces in the x, y directions (unchanged):

$$\sum F_x = Mg\sin\theta - f_s = Ma_{CM}$$
$$\sum F_y = N - Mg\cos\theta = 0$$

   - Sum of the torques about the point of contact (**use parallel axis theorem** for I!):

$$\sum \tau_z = -Mg\sin\theta R = \left(\frac{1}{2}MR^2 + MR^2\right)\alpha$$

3. Can immediately determine α (a_{CM}) in this case!

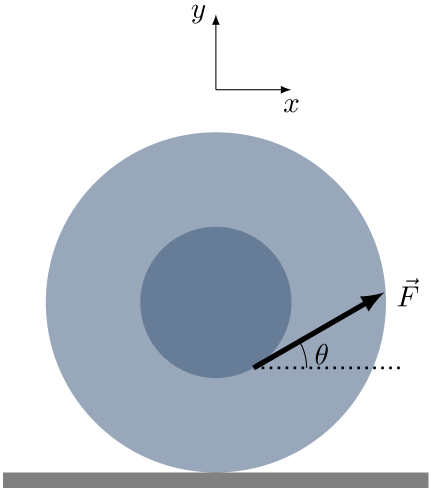


A cylinder rolling down an incline, modeled using instantaneous axis of rotation.

For rolling without slipping:

$$\boxed{a_{CM} = \alpha R}$$

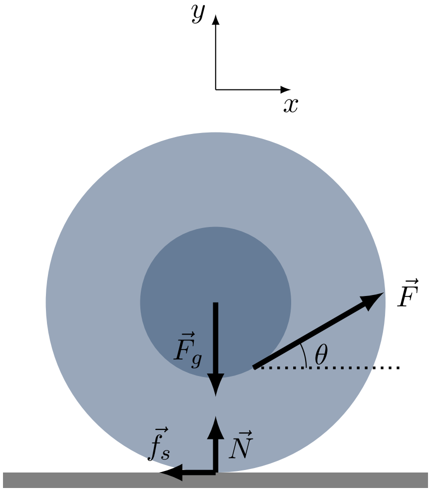
Example: Giant Yo-Yo



Modeling a giant yo-yo.

Example: Giant Yo-Yo

- The direction of motion of the yo-yo depends on the angle with which we pull the rope
- The direction of the force of friction is tricky to determine!
- When the angle is low:
  - The yo-yo goes towards the right (towards F).
  - $\vec{f}_s$ has to be towards the left so that the net torque about the CM is into the page. (The torque from F about the CM would make the wheel go to the left).



Modeling a giant yo-yo.

Example: Giant Yo-Yo

- Sum of forces:

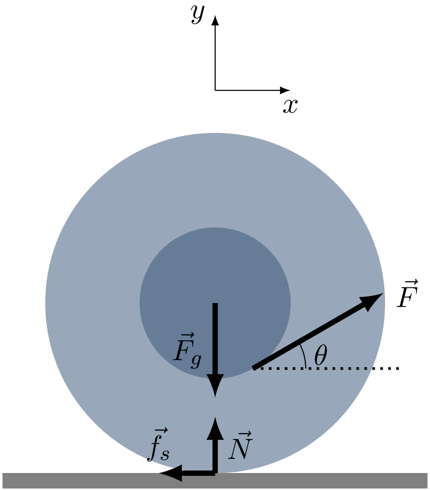
$$\sum F_x = F\cos\theta - f_s = Ma_{CM}$$
$$\sum F_y = F\sin\theta + N - Mg = 0$$

- Sum of torques **about CM** (only $\vec{F}$ and $\vec{f}_s$):

$$\sum \tau_z = FR_1 - f_sR_2 = I\alpha$$

Thinking about it

- When the angle is small, the torque about CM from $\vec{f}_s$ is larger than that from $\vec{F}$ (yo-yo goes to the right)
- Above some angle, the torque from $\vec{f}_s$ is less than that from $\vec{F}$ and the yo-yo goes the other way.
- There is an angle where the net torque is zero $\rightarrow$ the yo-yo just slides



Modeling a giant yo-yo.

Angular Momentum

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Rotational dynamics – angular momemtum

- Given an axis of rotation:

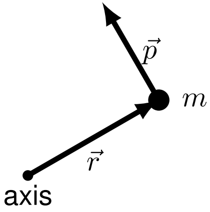
$$\vec{r} \times (\vec{F}^{net} = m\vec{a})$$
$$\therefore \vec{\tau}^{net} = mr^2\alpha$$

- Rotational version of momentum (angular momentum):

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}) = \cancelto{0 \quad (\vec{v} \parallel \vec{p})}{\frac{d\vec{r}}{dt} \times \vec{p}} + \vec{r} \times \frac{d\vec{p}}{dt} = \vec{\tau}^{net}$$

$$\therefore \boxed{\frac{d\vec{L}}{dt} = \vec{\tau}^{net}} \qquad \text{compare to} \qquad \frac{d\vec{p}}{dt} = \vec{F}^{net}$$



Angular momentum is defined for a particular axis of rotation based on the particle's momentum.

Angular momentum of particle

- If there is **no net torque** about an axis on the mass, then its **angular momentum relative to that axis is conserved**.
- There could be a net force on the particle (momentum is not conserved), but if the net force is in such a way that it produces no net torque (**about a given point**), then angular momentum (**about that given point**) is conserved.
- Angular momentum measures how much “rotation” a particle has about a particular axis.

$\rightarrow$ If the particle has no angular velocity about that axis, it has no angular momentum about that axis.

$\rightarrow$ If the particle is further from axis or more massive, it has more angular momentum.

In this case, linear momentum is not conserved while angular momentum about the given axis is conserved.

Angular momentum of a system

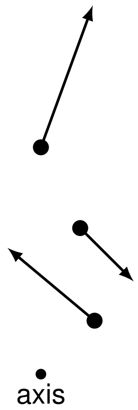
- We summed the momenta of the particles in a system:

$$\frac{d\vec{p}_i}{dt} = \vec{F}^{net}$$
$$\therefore \frac{d\vec{P}}{dt} = \vec{F}^{ext}$$

- We do the same for angular momentum:

$$\frac{d\vec{L}_i}{dt} = \vec{\tau}^{net}$$
$$\therefore \frac{d\vec{L}}{dt} = \vec{\tau}^{ext}$$

- If there is **no net external torque on a system**, the angular momentum of the system is conserved



The total angular momentum is the sum of the angular momenta of the particles in the system.

Angular momentum for a solid object

- For a single particle in a system its angular momentum about an axis of rotation is:

$$\vec{L}_i = \vec{r}_i \times \vec{p}_i = \vec{r}_i \times m_i \vec{v}_i = m_i \vec{r}_i \times (\vec{\omega} \times \vec{r}_i) = m_i r_i^2 \vec{\omega}$$

- Adding together the angular momenta of particles in a solid object (all with the same angular velocity):

$$\vec{L} = \left(\sum_i m_i r_i^2 \right) \vec{\omega}$$

$$\boxed{\vec{L} = I \vec{\omega}}$$

Correspondence between linear and angular

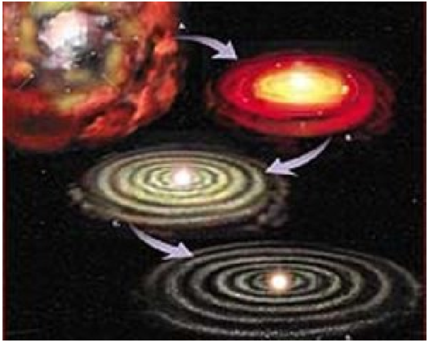
Name	Linear	Angular	Correspondence
Displacement	s	$\vec{\theta}$	$d\vec{\theta} = \frac{1}{r^2}\vec{r} \times d\vec{s}$
Velocity	$\vec{v}$	$\vec{\omega}$	$\vec{\omega} = \frac{1}{r^2}\vec{r} \times \vec{v}$, $v_T = \vec{\omega} \times \vec{r}$ [1]
Acceleration	$\vec{a}$	$\vec{\alpha}$	$\vec{\alpha} = \frac{1}{r^2}\vec{r} \times \vec{a}$, $a_T = \vec{\alpha} \times \vec{r}$ [2]
Inertia	m	I	$I = \sum_i m_i r_i^2$
Momentum	$\vec{p} = m\vec{v}$	$\vec{L} = I\vec{\omega}$	$\vec{L} = \vec{r} \times \vec{p}$
Newton's Second Law	$\vec{F}^{ext} = m\vec{a}_{CM}$	$\vec{\tau}^{ext} = I\vec{\alpha}$	$\vec{F} \to \vec{\tau}$, $m \to I$, $\vec{a} \to \vec{\alpha}$
Newton's Second Law	$\frac{d\vec{p}}{dt} = \vec{F}^{ext}$	$\frac{d\vec{L}}{dt} = \vec{\tau}^{ext}$	$\vec{F} \to \vec{\tau}$, $\vec{p} \to \vec{L}$
Kinetic energy	$\frac{1}{2}mv^2$	$\frac{1}{2}I\omega^2$	$m \to I$, $v \to \omega$
Power	$\vec{F} \cdot \vec{v}$	$\vec{\tau} \cdot \vec{\omega}$	$\vec{F} \to \vec{\tau}$, $\vec{v} \to \vec{\omega}$

[1] This corresponds to the component of velocity perpendicular to $\vec{r}$.

[2] This corresponds to the component of acceleration perpendicular to $\vec{r}$.

Planets and the solar system

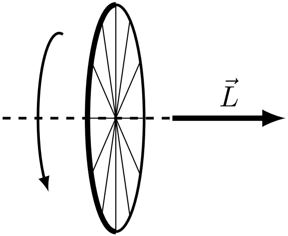
- The solar system started off as a very big roughly spherical ball of gas with atoms moving in all directions.
- Since the motion of the atoms was random, there was a net total angular momentum in some direction.
- Atoms and matter coalesced because of gravity towards the centre of mass:
  - Their rotational speed had to increase (moment of inertia got smaller).
  - They all ended up in the same plane (perpendicular to angular momentum of the system), mostly due to collisions.



Through conservation of angular momentum, all planet orbit in a similar plane. https://jcconwell.wordpress.com/tag/angular-momentum/

Example: Bicycle wheel

Tilt the axis of the wheel!

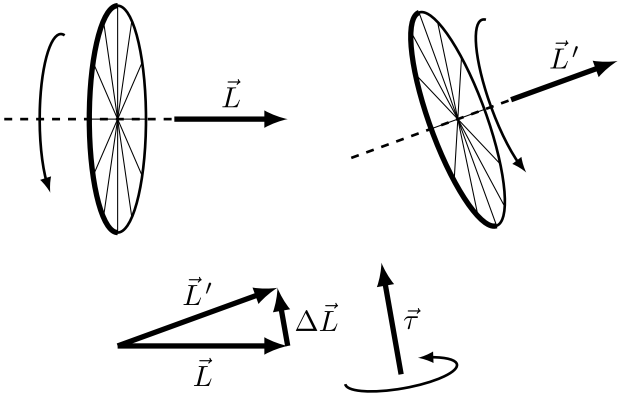


Tilting the bicycle wheel requires exerting a torque

- If you try to change the axis of rotation, you have to exert a torque to change the angular momentum.

$$\vec{\tau} = \frac{d\vec{L}}{dt} \sim \frac{\Delta\vec{L}}{\Delta t}$$

- If you let the wheel hang, gravity tries to rotate the axis of rotation $\rightarrow$ precession.



Tilting the horizontal axis of a spinning wheel requires a torque in the vertical direction!